MIL-C-50409(MU)
11 April 1969

MILITARY SPECIFICATION

COMPONENTS, METAL, FOR FIBER CONTAINER M105

1. SCOPE

1.1 This specification covers the hardened packing stop and packing stop support for M105A3 fiber container for 105MM Howtizer ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

FEDERAL

TEST METHOD STANDARD NO. 151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions.

FSC. - 8140

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

9243286 - Support, Stop, Packing
9243287 - Stop, Packing, Hardened

PUBLICATIONS

ARMY

ET-9243286 - Equipment Tabulation

(Copies of specifications, standards, drawing and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. Requirements

3.1 Material.-Material shall be in accordance with applicable drawing and specification.

3.2 Parts.-The parts shall comply with all requirements specified on Drawings (dwg) 9243286 and 9243287 and with all requirements specified in applicable specification.

3.3 Workmanship.-All parts shall be dry and free of chips, burrs, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor the parts be contaminated by the cleaning agent.

3.4 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of thirty-five packing stop supports and thirty-five hardened packing stops which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. Of the total submitted, a quantity of ten stops and ten supports shall be furnished hardened, but prior to plating. The balance shall be submitted fully finished in accordance with the applicable drawings and specifications. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Packing Support (see dwg. 9243286)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: Non defined			
Minor:	AQL 2.5 percent		
201.	Outside diameter	Gage	01001
202.	Inside diameter	Gage	01002
203.	Thickness	Gage	01003
204.	Finish	Visual	01004
205.	Evidence of poor workmanship (see 3.3)	Visual	01005

4.3.2.2 Hardened packing stop (see dwg. 9243287)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 2.5 percent		
201.	Outside diameter	Gage	02001
202.	Width of slot	Gage	02002
203.	Radius at back of slot	Gage	02003
204.	Thickness	Gage	02004
205.	Finish	Visual	02005
206.	Evidence of poor workmanship (see 3.3)	Visual	02006

4.3.3 Testing

4.3.3.1 Hardness of supports (see dwg. 9243286), Major defect, Code No. 01006. Ten hardened packing stop supports which have not been plated shall be randomly selected for this test. If any sample fails to comply with applicable requirements, the lot shall be rejected. The test shall be performed as specified in 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

4.3.3.2 Hardness of hardened packing stop (see dwg 9243287), Major defect, Code No. 02007. Ten hardened packing stops shall be randomly selected for this test. If any sample fails to comply with the applicable requirement the lot shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.3.3 Heat treatment of packing stop supports after plating (see dwg. 9243286), Major defect, Code No. 01007. Ten packing stop supports shall be randomly selected for this test. If any sample fails to comply with the applicable requirements, the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.4 Heat treatment of hardened packing stops after plating (see dwg. 9243287), Major defect, Code No. 02008. Ten hardened packing stops shall be randomly selected for this test. If any sample fails to comply with the applicable requirements, the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.5 Yield point of packing stop supports (see dwg. 9243286), Major defect, Code No. 01008. A test specimen shall be prepared for this test in accordance with the applicable specification. If the specimen fails to meet the applicable requirements, the production which it represents shall be rejected The test shall be performed as specified in 4.4.3.

4.3.3.6 Yield point of hardened packing stops (see dwg. 9243287), Major defect, Code No. 02009. A test specimen shall be prepared by this test in accordance with the applicable specification. If the specimen fails to meet the applicable requirements, the production which it represents shall be rejected. The test shall be performed as specified in 4.4.3.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-9243286 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspect on equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in a accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test Methods

4.4.1 Rockwell hardness test. The support and hardened stop shall be tested in accordance with Federal Test Method Standard No. 151, Method 243.1. The procedure shall be performed three times on each sample, and the hardness reported for the sample shall be the mean of the three results obtained from that sample. This test shall be performed for each lot or each eight hours production, whichever represents the smaller quantity of product.

4.4.2 Heat treatment after plating.-Ten plated samples shall be marked with a heat indicating substance, such as lacquer or paint, prior to final heat treatment. The indicating substance shall be chosen to have a temperature sensitivity of 350 degrees Fahrenheit, plus or minus three percent. The ten parts shall be inspected for the presence of the indicator. The samples shall then be subjected to final heat treatment together with the production lot which they represent. The ten samples shall be inspected to assure that the indicator gives evidence of having achieved its sensitivity temperature. This test shall be performed for each lot or each eight hours production, whichever represents the smaller quantity of product.

4.4.3 Yield point.-The packing stop supports and packing stops shall be tested in accordance with Federal Test Method Standard No. 151, Method 211.1. The test shall be performed for each lot or each eight hours production, whichever represents the samller quantity of product.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:24Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking

5.1 Level C - Packing and marking shall be in accordance with MIL-STD-1169.

NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data Cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1169.
c. First article inspection provision.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection eqquipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipmemt designs, or codes as indicated in paragraph 6.3.2 Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commericial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:24Z
Check the source to verify that this is the currentversion before use.

drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-Unless otherwise specified in EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND7. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:24Z
Check the source to verify that this is the currentversion before use.

and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 8140- A165

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:24Z
Check the source to verify that this is the current version before use.